A Low Energy Consuming MAC Protocol for Wireless Sensor Networks

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Abstract*—In a Wireless Sensor Network (WSN), sensor nodes are small autonomous devices that run on batteries. When the battery of a sensor node dies, it stops functioning. Generally, a sensor node loses more energy in communication than in sensing or processing activities. The proposed model, Low Energy Consuming Medium Access Control (LECMAC), minimizes energy consumption by reducing the number of communications in the network. Hence, the WSN lifetime increases. Two existing models, ES-MAC and LEACH, are compared to LECMAC in four various layouts. The experimental results show that LECMAC outperforms ES-MAC and LEACH in extending network lifetime.

Keywords*— Radio, Maximum Cluster Size, Cluster Head, Residual Energy.

I. Introduction

A Wireless Sensor Network (WSN) contains autonomous sensor nodes meant to screen physical or ecological conditions and send the data to a Base Station (BS) or gateway. A WSN is made up of a group of small sensor nodes that communicate with one another and share data[1]. A sensor node can perform its work as long as its source of power isn't depleted. A sensor node's death could have an impact on its neighbors, as they may be using the node to forward their data.

Each node has an onboard Sensor, Analog-to-digital converter (ADC), Processor (CPU), Radio (transceiver), Memory, and Power Source (Battery). Sensor nodes can be divided into different groups (clusters), each with a cluster head (CH), to save energy. The BS receives data from the CH, which it gathers from all of its members. Clustering makes sure that a majority of the transmissions are short.

Generally, a sensor node loses more energy in communication activities compared with processing or sensing activities [2]. The primary function of radio is transmitting and receiving control and data packets. When a sensor node is ready to transmit or receive packets, its radio is ON. This is referred to as Idle Listening. At times, the node may hear what its neighbors are sending or receiving. During their lifetime, sensor nodes spend an important amount of power listening to and overhearing in the background.

Collisions can occur when many sensor nodes attempt to send packets at the same time. All the packets that were corrupted in the collision must be retransmitted. This leads to a substantial amount of energy loss in sensor nodes. Although a capture effect may be used to recover certain packets, many conditions must be met for the recovery to be successful. Receiving packets meant for other nodes, a phenomenon is known as "overhearing.", is another cause of nonessential energy loss. Control-packet overhead is also a cause of unnecessary energy loss. Idle listening or listening to an empty channel with the hope of receiving traffic is one of the most common causes of energy loss. Over emitting, which occurs when a message is sent when the target node is not ready, also leads to energy loss in sensor nodes. MAC protocols that are designed for WSNs, should have inbuilt mechanisms to reduce energy loss due to these factors [3].

When it comes to CH selection, most MAC protocols follow a decentralized approach. To put it another way, a sensor node determines whether or not it should be designated as a CH in a particular round. However, this leads to a random distribution of CHs in the target area. Hence, comparatively speaking, some areas may be overrun by CHs compared to other areas. In addition, some clusters may grow much larger than others in terms of the number of nodes they contain. Again, this results in an uneven distribution of the network's CHs' workloads [4].

The work presented in this paper, Low Energy Consuming Medium Access Control (LECMAC), extends the network lifetime by minimizing energy loss due to some of the factors discussed above. The model usesfiveparameters:Proximity Threshold (PT), Maximum Cluster Size (MCS), Threshold Energy (TE), Distance-to-Event (DE) Verification Period (VP). The TE makes sure that sensor nodes with small energy are excluded from the CH selection process. The PT makes sure no region in the target area is crowded with CHs. The MCS ensures no cluster is overpopulated with member nodes. The VP is used to ensure that the idle listening in CHs is reduced. The DE is used to minimize the transmission of redundant data.

The rest of the document is organized as follows. Section II examines related research. Section III explains the suggested model. Section IV defines the simulation results. Finally, section V concludes the paper.

II. Related work

There are two basic MAC schemes: Carrier Sense Multiple Access (CSMA) and Time Division Multiple Access(TDMA). Nodes in a CSMA-based MAC network compete for medium

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access with each other. In a TDMA-based MAC scheme, each node is allocated time slot(s) for sending and receiving packets. CSMA-based MAC protocols are significantly more flexible and have lower latency than TDMA-based MAC protocols, and they are also significantly more cost-effective.Because of the increased idle listening, overhearing, and network collisions, CSMA-based MAC protocols are more energy-consuming. Some fundamental TDMA based MAC protocols are examined below.

In LEACH [5], for every node in the network, a random number (RN) in the range of 0 to 1 is generated and compared to a threshold energy value. In the current round, a node has to become a CH if its RN is lower than the threshold. ADV packets are sent by the CH to all non-CH nodes within its transmission range, and they are all received. A non-CH may receive multiple ADV packets but chooses to join the CH the strongest signal (ADV) was received from. The non-CH sends a JOIN-REQ to the CH. After receiving JOIN-REQ messages, A CH telecasts a transmission schedule so that its members can get their own data slots. A member can transmit only in its data slot. To avoid interfering with other members, the radio is disabled in their data slots. The BS receives data that a CH collects from its members.

In ES-MAC [6],a CH switches its radio ON at the starting of each frame for a very short duration (VP). The radio is turned off if there is no incoming data packet during the VP. The CH will keep its radio on until the end of the slot if it detects an incoming data packet. ES-MAC reduces CH idle listening compared to LEACH.

In BMA-MAC [7],the member nodes are allocated short time slots called control slots to send control packets to the CH. The purpose of sending a control packet is to inform the CH that the identical node has data to communicate. A node transmits a control packet only if it has data. If a node does not have any data, it will stay idle. A CH allocates data slots only to those members that send control packets.

In LDC-MAC [8], each node receives a data slot and a control slot when a CH broadcasts a transmission schedule. The CH assumes that a component node that has sent a control packet will also send a data packet. These nodes' data slots require the CH to keep its radio on.

In BEE-MAC [9], when the control period begins, a source node turns on its radio to listen for control packets from other source nodes. The node uses this data to determine which slots are available and which have been claimed. If a node learns that the first six data slots have been claimed, it will claim one or more data slots starting from the seventh data slot. Unlike LDC-MAC, all the unused slots in BEE-MAC are located towards the frame's end.

Some of the issues in the MAC protocols discussed above are addressed by LECMAC. The TE value is used to keep low-energy nodes out of the cluster head selection process. The PT value is used to prevent CHs from crowding in any region of the target area. Clusters can't get too big if the MCS value is set high enough. The VP value is used to reduce the amount of time spent listening during CHs. The DE value is used to reduce the transmission of redundant data. The proposed model is detailed in the following section.

III. Proposed Model

Beginning of a round, the Residual Energy (RE) of each sensor node is checked against the TE. Participation in the CH selection procedure is allowed only for nodes with RE greater than TE. Such a node is a potential CH and competes with its neighbors to telecast an ADV packet. Node B (A's neighbor) checks if node A is within B's PT if node A telecasts an ADV packet first (predefined value in meters). If so, node B turns its status from a potential CH to a non-CH node and does not broadcast an ADV packet. If node A is not within node B's PT, B will continue trying to broadcast an ADV packet.

After a delay period, each CH sends out an invitation (INV) message to non-CH nodes within its transmission range to extend an invitation to them to participate in the cluster. Multiple INV messages may be received by a non-CH. It records the CH IDs in all the received INV messages. The non-CH node sends a JOIN-REQ message to the CH the strongest signal (INV) was received from (assumed to be the closest). A CH will accept requests as long as the MCS is not reached. If the MCS is reached, the CH broadcasts a REJECT message to decline requests of the remaining non-CH nodes. A rejected non-CH node sends a JOIN-REQ message to the next closest CH an INV message was received from. In this way, a non-CH node continues its attempt to join a cluster until it is successful.

After cluster generation, a CH telecasts an intra-cluster transmission schedule. Each member is assigned a data slot in a frame. The DE (predefined value in meters) ensures that only the node(s) closer to an event report the event. If a node senses an event more than DE meters away, it will not report the event to the CH. Hence, LECMAC can minimize the transmission of redundant data. A source node can transfer data only in itsaccepted slot. It remains its radio OFF in every other slot. A CH collects data from the source nodes, aggregates it, and sends the aggregated data to the BS. After n frames, a new round begins with the same process.

IV. Results and Analysis

Two existing models, ES-MAC and LEACH, are compared to the proposed model to determine which model is more efficient in energy conservation. The energy consumption analysis uses the First Order Radio Model. Every model is simulated with the help of the Python programming language. The following expectations are created for the simulation of all the protocols.

- Nodes are scattered throughout the target area randomly.
- Nodes are homogeneous and stationary.
- Nodes can communicate with the BS directly.
- Nodes can increase or decrease their transmission power as required.
- The BS has an infinite source of energy and is located outside of the target region.

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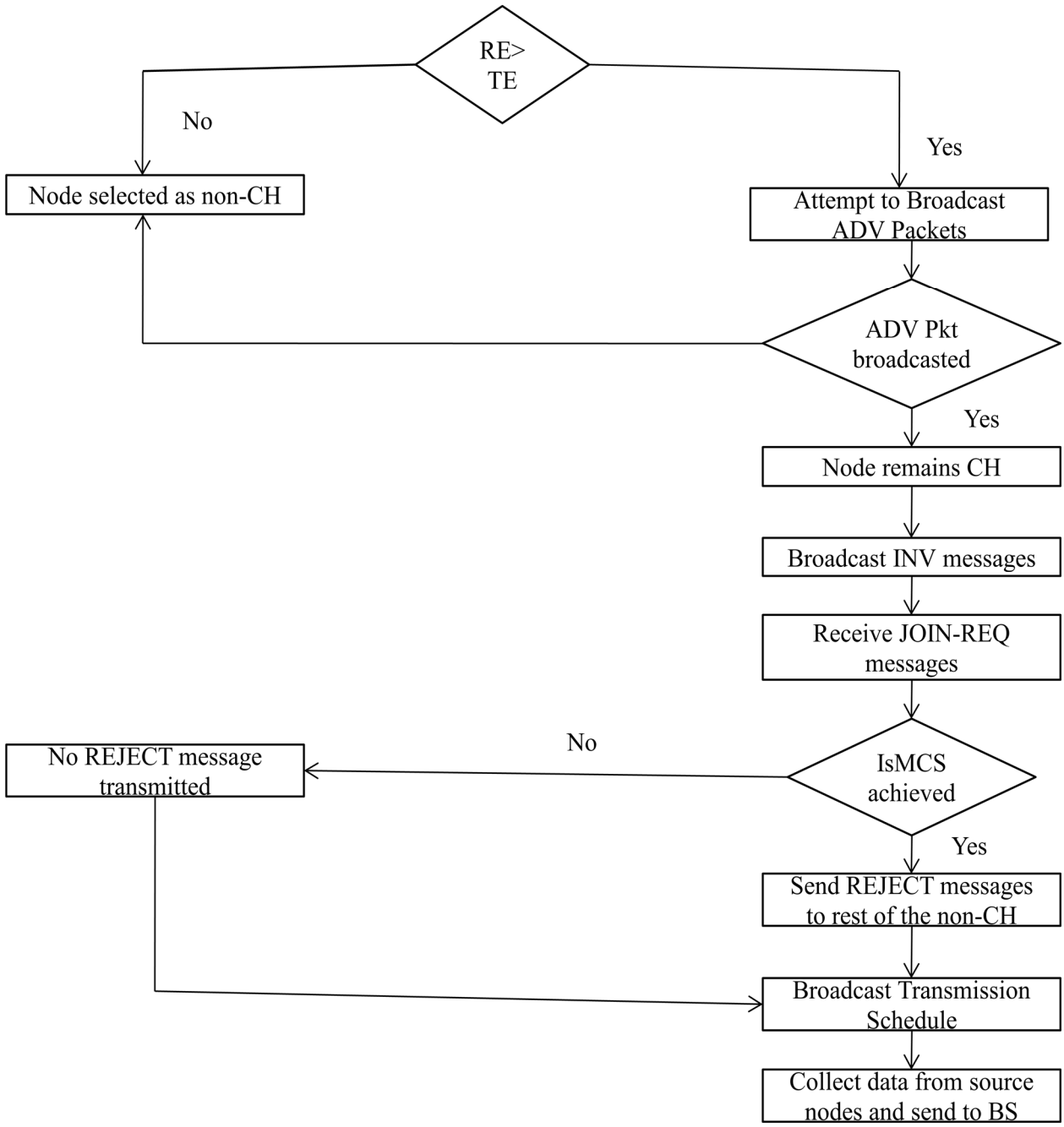

Fig. 1: Flow Chart of Proposed Model

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TABLE 1: PARAMETERS FOR SIMULATION

Parameters	Value
Threshold Energy	0.1 Joules
Target area	$100\times100m^2/200\times200m^2$
Number of nodes	100/200
Node initial energy	5 Joules
Proximity threshold	10m
E_{elec}	50 nJ/bit
Data/schedule packet size	100 bytes
E_{amp}	100 pJ/bit/m^2
Maximum Cluster size	15 nodes
Control packet size	20 bytes
E_{idle}	40 nJ/bit

Table 1 represents the parameter used in the simulation. Here E_{elec} is the energy used in transmitting or receiving a signal. The term E_{idle} refers to the amount of energy used while in idle mode and E_{amp} measures the voltage used by the transmit amplifier.

In the first layout, 100 nodes are placed in a 100×100 m^2 region. Fig. 2 shows the outcomes of the simulations in the first layout. Table 2 shows the first and last nodes dying in each simulation.

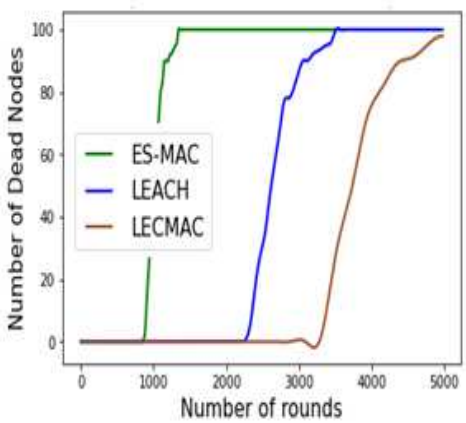

Fig. 2: Result in the First Layout

TABLE 2: THE LIFETIME OF WSN IN THE FIRST LAYOUT

Technique	First Node dies in RN	Last Node dies in RN
ES-MAC	821	970
LEACH	2272	4080
LECMAC	3269	4890

In the second layout, 200 nodes are placed in a 100×100 m^2 region. Fig. 3 shows the outcomes of the simulations in the second layout. Table 3 shows the first and last nodes dying in each simulation.

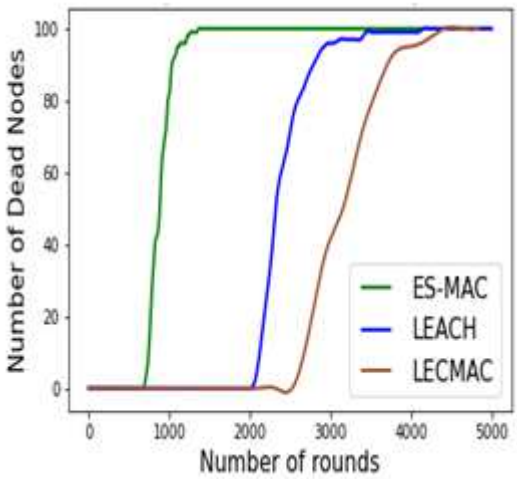

Fig. 3: Result in the Second layout

TABLE 3: THE LIFETIME OF WSN IN THE SECOND LAYOUT

Technique	First Node dies in RN	Last Node dies in RN
ES-MAC	718	1579
LEACH	1890	3200
LECMAC	2673	4685

In the third layout, 100 nodes are deployed in a 200×200 m^2 region. Fig. 4 shows the outcomes of the simulations in the third layout. Table 4 shows the first and last nodes dying in each simulation.

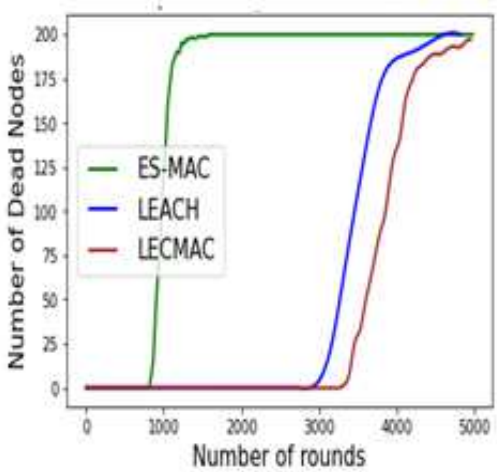

Fig. 4: Result in the third layout

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TABLE 4: THE LIFETIME OF WSN INTHIRD LAYOUT

Technique	First Node dies in RN	Last Node dies in RN
ES-MAC	681	1360
LEACH	2064	4090
LECMAC	2527	4250

In the fourth layout, 200 nodes are placed in a 200×200 m^2 region. Fig. 5 shows the outcomes of the simulations in the fourth layout. Table 5 shows the first and last nodes dying in each simulation.

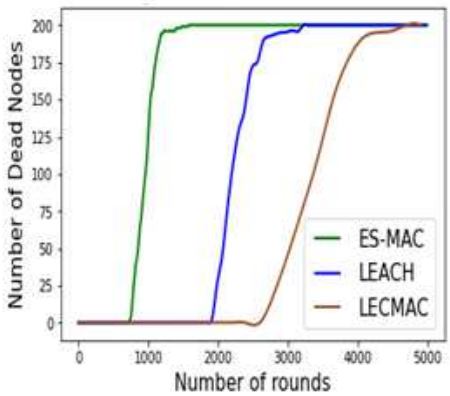

Fig. 5: Result in the Fourth layout

TABLE 5: THE LIFETIME OF WSN INTHE FOURTH LAYOUT

Technique	First Node dies in RN	Last Node dies in RN
ES-MAC	790	1618
LEACH	1975	3329
LECMAC	2732	4750

According to the simulation results, LECMAC is better than ES-MAC and LEACH in terms of energy conservation in sensor nodes. The proposed model reduces energy usage in sensor nodes by (i) not including low-energy sensor nodes in the CH selection process, (ii) not allowing any specific region in the target area to be crowded with CHs, (iii) not allowing any cluster to grow too big compared with other clusters, (iv) reducing the idle listening in CHs, and (v) minimizing the transmission of redundant data.

V. CONCLUSION

Sensor nodes rely on batteries to perform their operations. Therefore, MAC protocols for WSNs should be designed to reduce energy consumption in sensor nodes. The suggested model resolves some standard issues in WSNs and extends the network lifetime of WSNs. This study compares the proposed model, LECMAC, with ES-MAC and LEACH in four different layouts. The simulation findings demonstrate that LECMAC performs significantly better than ES-MAC and LEACH in dealing with some standard energy-consuming factors. Other WSN issues like redundant data transfer and long-range communications can be added to the model.

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